

Capitol Hill Crawler: Technical Manual V9 (Executive & 128-Pool Integration)

1. Systemic Architecture: The Polarization Engine

1.1 The Polarization Scale

The Capitol_Hill_Crawler core engine utilizes a bounded continuous numerical array 0, 100 to define the ideological landscape. This scale functions as a global balancing modifier that dictates NPC aggression, dialogue branching, and interface difficulty:

- **Left-Leaning / Progressive:** 0, 49
- **Bipartisanship Equilibrium:** 50
- **Right-Leaning / Conservative:** 51, 100

1.2 Mathematical Difficulty Delta

Programmatic encounter difficulty is a function of the ideological distance between the player's legislation and the target node. With the **Base Bill Value (B_v)** fixed at 50 for Tier 1 operations, the system calculates the **Difficulty Delta (D)** using the following formula: $D = |B_v - N_r|$ Where N_r represents the polarization rating of the **Target Representative Node**.

1.3 UI Interface Constraint Table

The Difficulty_Delta (D) determines the specific UI_Interface_Mode. For values where $D \geq 40$, the system identifies the node as "Fringe" (Extremist) relative to the B_v equilibrium of 50.

Difficulty Delta (D)	Political Context	UI Interface Mode	Mechanic Constraints
$0 \leq D \leq 5$	Intra-Party (Ally)	3-Option Multiple Choice	Standard timing; clear strings.
$5 \leq D \leq 15$	Cross-Aisle (Opposition)	Open-Ended Keyboard Input	Clear text; dialogue-based context hints.
$D \geq 40$	Intra-Party (Fringe)	12-Option High-Confusion Matrix	Semantic similarity distractors used to penalize guessing. Cross-Aisle (Fringe) Mumbled/Scrambled Input Procedural text degradation; crucial keywords omitted via "..." strings.

2. Global Civics Database: The 128-Question Pool

2.1 Database Schema & Categorization

The Civics_Database synthesizes the 128 questions from the 2025 version of the U.S. Civics Test. For implementation, these are grouped into the following technical categories:

Category_ID	Theme	Question Range	Total Count
CAT_01	American Government	Q1 – Q72	72
CAT_02	American History	Q73 – Q118	46
CAT_03	Symbols and Holidays	Q119 – Q128	10

2.2 The "Scholar's Path" Bypass Logic

The engine monitors player metadata for the **65/20 Special Consideration** criteria (Age $\geq$ 65 AND Residency $\geq$ 20 years). If Scholar_Path_Active is TRUE:

1. The Question_Pool is restricted **exclusively** to the 20 asterisked (*) questions found at the end of each section in the source test.
2. The engine selects a 10-question array from this restricted pool.
3. The Success_State threshold is set to 6 correct answers (60%).

2.3 High-Confusion Entry Logic

For high-polarization encounters (≥ 40), the engine populates a 12-option matrix. A primary example is MC_02 (Question 83: The Federalist Papers). Distraction logic mandates the inclusion of semantically similar historical figures (e.g., Jefferson, Adams, Franklin) alongside the four correct source answers (Madison, Hamilton, Jay, Publius) to ensure high-accuracy requirements.

3. House and Senate Progression Logic

3.1 The 175-Vote Extremist Ambush

The global engine constantly monitors the Vote_Counter. When Vote_Counter ≥ 175 , the system locks standard movement and triggers the Extremist Ambush. Logic branches based on the Committee_Status flag:

- **Friendly Markup Mode (Success):** Triggered by correct input. The Committee_Status is set to SUCCESS, converting the Extremist node into a temporary modifier asset that bypasses obstacles in the Committee mini-dungeon.
- **Hostile Markup Mode (Failure):** Triggered by incorrect or timed-out input. Committee_Status is set to FAILURE. The Committee Room instantiates as an aggressive gauntlet where the player must clear micro-negotiations with individual backbenchers.

3.2 218-Vote Majoritarian Threshold

To clear the House Wing and unlock the Rotunda Hub transition, the Vote_Counter must reach a minimum integer value of **218** .

3.3 Senate Floor Gauntlets

Senate progression requires achieving a **60-vote Cloture threshold** .

- **Filibuster Gauntlet:** Instantiated at Senate_Vote_Counter = 50. The player must clear 5 rapid-fire Multiple Choice questions with 4 options each. The pause function is disabled during this sequence.
- **Overtime Fail Loop:** Failure in the gauntlet triggers the Gridlock environment flag. The exit is hard-locked. The player must navigate hidden cloakroom maps to locate **10 Rogue Senator nodes** and solve open-ended challenges to secure the final 10 votes.

4. Updated INF Economy and Rotunda Shop

4.1 Currency Generation

The Influence_Points (INF) variable serves as the primary currency. Per V9 Directive, **each correct answer increments the Influence_Points total by 100 units.**

4.2 Standardized Shop Pricing

All items at the Rotunda Shop vendor are set to a flat cost of **300 INF**.

4.3 Item Mechanics

- **Talking Points Memo:** Programmatically strips 3 incorrect choice strings from any active Multiple-Choice grid.
- **Hearing Aid:** Identifies and replaces scrambled "..." strings in mumbled extremist questions with their plaintext target keywords.
- **Cafeteria Coffee:** Suspends the execution thread for the "Point of Order" confidence timer in the Senate, effectively freezing the clock.
- **Super PAC Injection:** Sets a Bypass_Flag that automatically clears 5 required Rogue Senator nodes during the Senate Overtime loop, allowing for faster Cloture.

5. Final Stage: The Presidential Interview

5.1 Ally President Encounter Mechanics

- **Initialization:** Loop of 20 Multiple Choice questions.
- **Modifier Interaction:** Default is 4 options; reduced to 3 if a Talking_Points_Memo is active.
- **Passing Threshold:** Correct_Answers $\geq$ 12 required to set the Bill_Signed flag to TRUE.

5.2 Opposition President Showdown Mechanics

- **Initialization:** Loop of 10 Open-Ended keyboard input questions.
- **Timer Constraint:** Hardcoded 30-second ticking clock per question block.
- **Fuzzy Logic Check:** To account for mechanical keystroke errors on Chromebook hardware, the compiler must use a **Levenshtein edit distance of $\leq$ 2** for string matching.
- **Passing Threshold:** Correct_Answers $\geq$ 6 required to override the Veto and trigger the success state.

6. Scoring and JSON State Export

6.1 High Score Algorithm

The final efficiency score is calculated as follows:
$$Final_Score = Base_Clear_Points + (Unanswered_Questions \times 150)$$

6.2 State Saving

The `Save_Station` terminal in the Rotunda exports a JSON string to local storage for persistent progression. **Sample JSON Snippet:**

```
{
  "Session_ID": "TC-9982",
  "Vote_Counter": 218,
  "Influence_Points": 1200,
  "Committee_Status": "Friendly_Markup",
  "Passed_Flags": {
    "House_Cleared": true,
    "Senate_Cloture": false,
    "Scholar_Path_Active": false
  },
  "Notebook_Entries": ["HB_435_VOTING_MEMBERS", "SB_CLOTURE_60"]
}
```